

LESSON 1: Git

What is git?

Git is a free, open-source distributed version control system designed to track changes in source code and files over time. It allows multiple developers to work on the same project simultaneously without overwriting each other's work. Created by Linus Torvalds in 2005 for Linux kernel development, Git has become the global standard for modern software engineering.

Essential Git Commands

To use Git via the command line, developers rely on a few foundational commands:

- **git init**
- **git status**
- **git add <file>**
- **git commit -m "message"**
- **git push**

1. Git init

What is Git init?

git init is the command used to create a new, empty Git repository or reinitialize an existing one. Running this command transforms your current directory into a Git repository, allowing you to start tracking changes, managing code versions, and creating development branches.

How to use Git init?

To use git init, open terminal, cd (your project folder name), and run the command git init

Example:

```
student@teacher-Precision-T1700: ~/PROJECT-NAME
student@teacher-Precision-T1700:~$ cd PROJECT-NAME/
student@teacher-Precision-T1700:~/PROJECT-NAME$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/student/PROJECT-NAME/.git/
student@teacher-Precision-T1700:~/PROJECT-NAME$
```

2. Git add

What is git status?

git status command displays the current state of your Git working directory and staging area. It serves as an informative snapshot that tells you which branch you are working on, which files have changed, and whether those changes are ready to be saved.

How to use git status?

1. Open terminal
2. Use command “cd (your project name)”
3. Use command git status

Example:

```
student@teacher-Precision-T1700: ~/PROJECT-NAME
student@teacher-Precision-T1700:~$ cd PROJECT-NAME/
student@teacher-Precision-T1700:~/PROJECT-NAME$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    log_qstopmotion.txt

nothing added to commit but untracked files present (use "git add" to track)
student@teacher-Precision-T1700:~/PROJECT-NAME$
```

3. git add

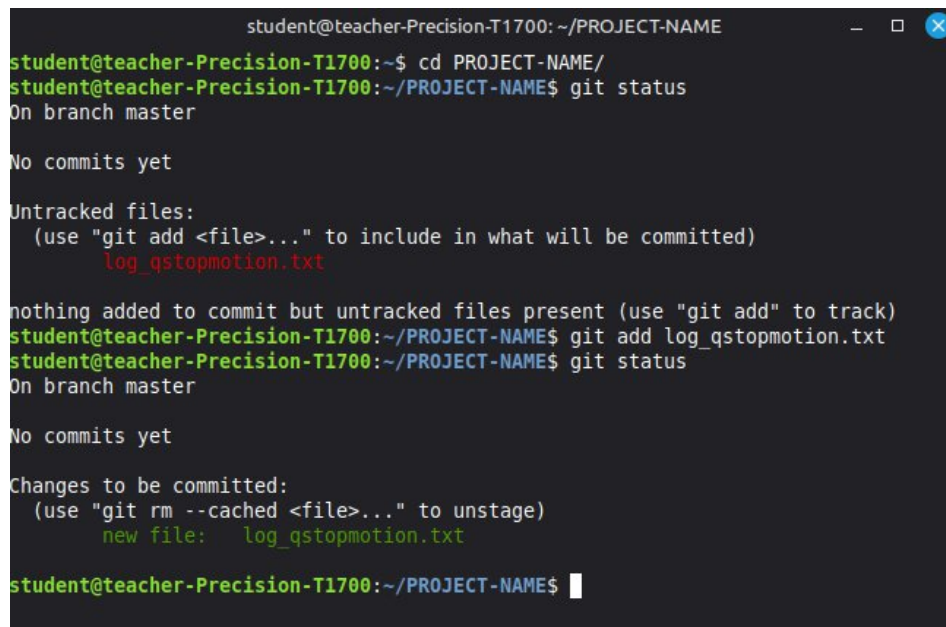
What is git add?

git add command moves changes from your local working directory to the Git staging area (also called the "index"), preparing them to be included in your next commit.

How to use git add

1. Open terminal
2. Use command "cd (your project name)"
3. Use command "git status" to see your file that you need to add
4. Use command "git add (file name)"
5. Use git status again and you will see you file turn green

Example:

A terminal window titled 'student@teacher-Precision-T1700: ~/PROJECT-NAME' showing the execution of git commands. The user navigates to the project directory and runs 'git status', which shows 'log_qstopmotion.txt' as an untracked file. Then, the user runs 'git add log_qstopmotion.txt', and a subsequent 'git status' shows the file as 'new file: log_qstopmotion.txt' ready to be committed.

```
student@teacher-Precision-T1700: ~/PROJECT-NAME
student@teacher-Precision-T1700:~$ cd PROJECT-NAME/
student@teacher-Precision-T1700:~/PROJECT-NAME$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        log_qstopmotion.txt

nothing added to commit but untracked files present (use "git add" to track)
student@teacher-Precision-T1700:~/PROJECT-NAME$ git add log_qstopmotion.txt
student@teacher-Precision-T1700:~/PROJECT-NAME$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   log_qstopmotion.txt

student@teacher-Precision-T1700:~/PROJECT-NAME$
```

4. git commit -m "message"

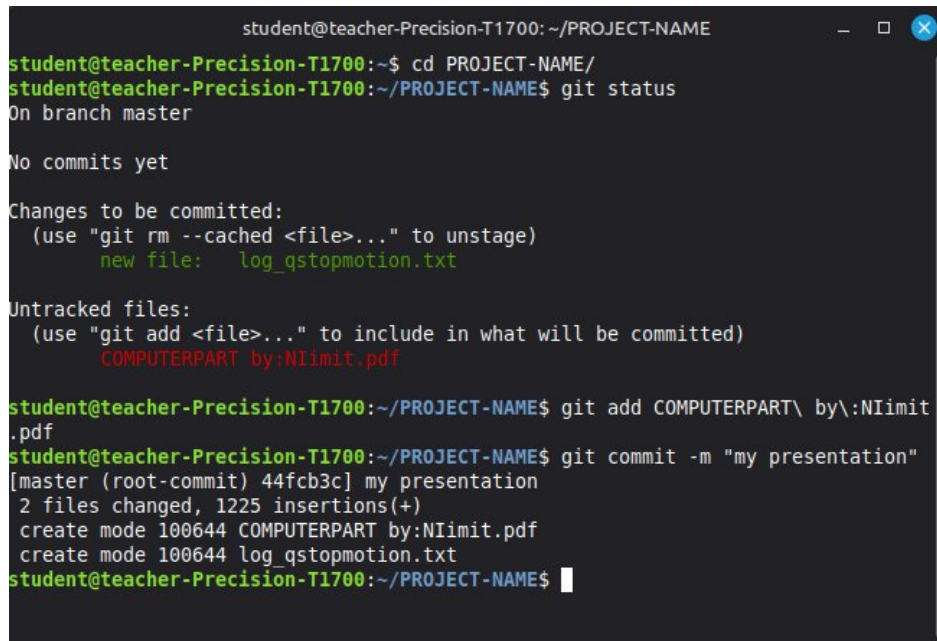
What is git commit?

git commit is a command that captures a snapshot of your project's staged changes and saves it permanently to your local repository history. Think of it like a save point in a video game; if you make a mistake later, you can always revert your code back to this exact moment.

How to use it

1. Open terminal
2. Use command "cd <your file name>"
3. Use command "git commit -m "your message" after use command git add"

Example:

A terminal window titled 'student@teacher-Precision-T1700: ~/PROJECT-NAME' showing the following commands and output:

```
student@teacher-Precision-T1700:~$ cd PROJECT-NAME/
student@teacher-Precision-T1700:~/PROJECT-NAME$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   log_qstopmotion.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        COMPUTERPART by:NIimit.pdf

student@teacher-Precision-T1700:~/PROJECT-NAME$ git add COMPUTERPART\ by:NIimit
.pdf
student@teacher-Precision-T1700:~/PROJECT-NAME$ git commit -m "my presentation"
[master (root-commit) 44fcb3c] my presentation
 2 files changed, 1225 insertions(+)
 create mode 100644 COMPUTERPART by:NIimit.pdf
 create mode 100644 log_qstopmotion.txt
student@teacher-Precision-T1700:~/PROJECT-NAME$
```

5. git push

What is git push?

git push is a command-line utility used to upload local repository commits to a remote repository. It essentially publishes your local code changes to a shared server like GitHub, GitLab, or Bitbucket so other collaborators can access and pull your updates.

How to use it

1. Open terminal
2. Use command "cd <your file name>"
3. Use command "git push" and then enter your username and your password after commit your file

Example:

```
student@teacher-Precision-T1700:~$ cd project-Nimit/
student@teacher-Precision-T1700:~/project-Nimit$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    .~lock.git.docx#
    ach ach.docx
    git.docx

nothing added to commit but untracked files present (use "git add" to track)
student@teacher-Precision-T1700:~/project-Nimit$ git add git.docx
student@teacher-Precision-T1700:~/project-Nimit$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   git.docx

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    .~lock.git.docx#
    ach ach.docx

student@teacher-Precision-T1700:~/project-Nimit$ git commit -m "your message"
[main 489441f] your message
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 git.docx
student@teacher-Precision-T1700:~/project-Nimit$ git push
Username for 'https://github.com': Kulen-student
Password for 'https://Kulen-student@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 172.88 KiB | 28.81 MiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Kulen-student/nimit-repo.git
   83d0287..489441f  main -> main
student@teacher-Precision-T1700:~/project-Nimit$
```